

entire exercise was demonstrated live at the SEMSIM symposium in 2015.

Threye is now working on using a fixed wing drone, and upgrading the camera. We asked Anurag Rana, founder and CEO of Threye, about the advantages of using fixed wing drones over quadcopters, and a different camera. "Fixed wing will lead to higher endurance, longer flying time and doing more surveying per charge of take-off and landing," Rana said, then elaborates on the planned camera upgrades, "we did with a simple go-pro. the current state of the art is to use mirror-less DSLRs."

INTELLIGENCE, SURVEILLANCE AND RECONNAISSANCE

At the Aeronautical Test Range (ATR) near Chitradurga, the DRDO is testing the capabilities of a flying platform known as the Tactical Airborne Platform for Aerial Surveillance (TAPAS), initially known as Rustom. The platform is similar to the infamous Predator drones used by the US military. The Rustom-I can operate up to 15,000 feet above sea level, with an endurance of 12 hours. The upgraded Rustom-II can operate at even higher altitudes – 30,000 feet above sea level, and has an endurance of 24 hours. Primarily designed to carry a synthetic aperture radar payload, the drone is for intelligence, surveillance and reconnaissance (ISR) roles for all three armed forces. The drone also has the capabilities of carrying various other payloads with a total weight of up to 350 kg. The DRDO is indigenously developing suitable payloads for the drone. In the future, the Tapas platform has the potential to be weaponised. The DRDO Nishant is also been developed for IRS purposes, and has a range of four and a half hours. At the moment, the drone has to be launched from a pneumatic platform, but DRDO is modifying the drone to allow conventional take offs and landings. This modified version of Nishant is called Panchi. However, being capable of launching without an airstrip, and a parachute based recovery option, are both assets in themselves. The DRDO is also actively developing a combat capable drone, a secretive project known as Aura. The

platform is still in the project definition phase, but is expected to be a stealth drone that will be capable of launching missiles.

The Aquilon Mini UAV is a micro fixed wing drone that can be launched by hand. The entire operation, from the launch to the landing, can be autonomous. The Aquilon has automatic target acquisition and targeting capabilities. With electric propulsion, the UAV has a low noise signature, low visibility, low operational footprint. These properties make it ideal for covert operations.

UAVs are being used for augmenting the capabilities of the existing troops in various ways. The Indian military is in the process of procuring

THE FUTURE OF WARFARE IS AUTONOMOUS

If you are interested in the many lethal autonomous weapons systems (LAWS) being developed around the world, then the Digit Geek site has a deep dive into



the topic. It is not just aerial drones that are covered, but also ground based tactical and support units, armoured vehicles, surveillance ships, and even classified underwater submarines loaded with nuclear weapons. <https://dgit.in/laws>

mini UAVs to enhance the surveillance and situational awareness capabilities of infantry battalions. Cyient Solutions, a Hyderabad based company that is a joint venture between Cyient Ltd and Israel based BlueBird Aero Systems has bagged a supply order for UAVs to be used in extreme conditions. The SpyLite drones are suited for real time visual intelligence, and was the only drone to pass a demanding test process in high altitude extreme weather conditions.

TARGET PRACTICE

Another interesting use of drones is for target practice for other weapons. These may be aerial combat aircraft, or ground based anti aircraft systems. The Lakshya is one such Pilotless Target Aircraft (PTA), and can be launched from a sea vessel or the ground. It can dive from 800 meters to just 12 meters, and maintain a particular altitude for weapons demonstrations. The Ulka is another target drone used for testing missiles, and is launched from supersonic aircraft.

One of the companies working closely with the military for developing fixed wing UAVs is Tata Advanced Systems. Mini and micro UAVs are indigenously designed and developed for ISR applications. The company also develops the necessary support infrastructure and subsystems, such as ground control station, autopilot and gimbals for tracking.

The DRDO is also involved in developing the payloads for any current and future UAVs, with the potential to enhance defence capabilities. One such sub system is the Synthetic Aperture Radar (SAR), which is an airborne sensor for surveillance and reconnaissance. The SAR is designed to be capable of tracking slow moving targets such as convoys, trucks and armoured vehicles. The system has been tested for over 60 hours on 40 sorties.

Right now, the defence forces have limited capabilities of identifying, tracking and taking down UAVs. The Indian Air Force is in the process of inducting the Spyder LLQRM System, a surface to air missile defence system that has the capabilities of taking down hostile UAVs. The Spyder system is also capable of taking down enemy planes, helicopters and missiles at the same time.

Another cool concept that DRDO is working on, is the use of high energy lasers to take down hostile drones. The 10kw proof of concept system is known as the Experimental Technology Modules for Directed Energy Laser System. A tracking gimbal keeps the energy beam focussed on a moving target. For this system, tests have been conducted on an off the shelf quadcopter at a range of 1km. 